

Assessment and Analysis of Wearables and Companion Mobile (Health) Applications: A Usability Evaluation Framework

Preetha Moorthy

Department of Biomedical Informatics, Center for Preventive Medicine and Digital Health, Medical Faculty Mannheim,
Heidelberg University

preetha.moorthy@medma.uni-heidelberg.de

ABSTRACT

Wearable technologies have gained significant traction in various domains with healthcare being a particularly prominent area for their use, often in conjunction with mobile (health) applications. The increasing demand and versatility of these technologies pose challenges in meeting the specific needs of patients, healthy subjects, as well as healthcare professionals. Despite their growing adoption, there is a notable lack of empirical research focusing on user behavior, adaptation, acceptance, and usability evaluation for wearable technologies and companion apps. Research focusing on human factors and usability is currently inadequate in this specific area. Therefore, the objective of my research is to develop a structured guideline to address these limitations. The primary goal is to create a practical and flexible framework for assessing the usability of wearables and their associated mobile apps. This framework will be designed to be accessible and adaptable, allowing both usability experts and non-experts such as clinical researchers to evaluate wearable technology together with its companion mobile app. Through the development, validation, and implementation of this framework, this investigation aims to support the evaluation in a structured manner, and thus enhance the long-term adoption and improve the acceptance of wearable technologies and mobile (health) apps.

CCS CONCEPTS

• **Human-centered computing**; • **Ubiquitous and mobile computing**; • **Ubiquitous and mobile computing design and evaluation methods**;

KEYWORDS

Wearables, Mobile Health Applications, Usability Attributes, Evaluation Framework, Healthcare

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1 PROBLEM STATEMENT

Within the medical sector, healthcare professionals and clinical researchers prefer selecting wearables and mobile (health) apps for clinical studies, customizing them to their specific medical domains. Their proficiency in evaluating the usability of these tools is often significantly restricted, resulting in a limited understanding of the technology's usability and associated issues. As the demand for wearable technologies in healthcare continues to increase at a rapid pace, the evaluation of these devices in clinical settings becomes imperative to ensure the safety and well-being of patients. Therefore, the identified problem statements in this context are as follows:

Problem I: There is a significant gap in understanding the concepts of usability and user experience, especially in the healthcare field. Usability, often considered as a subset of user experience remains ambiguous, especially in the development of medical devices, including wearable technologies and mobile applications (mobile apps) [47, 74].

RQ 1: What usability attributes, methods, and frameworks have been most frequently measured/ evaluated when testing wearables and mobile (health) applications?

This research question aims to investigate the evaluation of wearables and mobile apps and collect and analyze currently existing approaches.

Problem II: Usability and user experience design guidelines created by usability practitioners are widely available mostly for the design and development of wearables as well as mobile apps. However, within the healthcare domain, there is a lack of comprehensive guidelines for conducting usability and user experience evaluations that specifically target wearables and their companion mobile (health) app, ensuring their fitness for purpose, ease of use, and wearability [4, 38, 54].

RQ 2: How can we better support the usability evaluation of wearables and their companion mobile (health) applications?

This research question aims to address shortcomings of usability evaluation of wearables and their companion mobile apps through the construction of an evaluation framework, based on the synthesis of existing approaches.

Problem III: In the healthcare domain, clinicians and researchers often select and customize consumer wearables and mobile (health) apps to align with their specific medical domains, intended use cases, and available technological features. However, these groups have limited proficiency in evaluating the usability of these technologies and often lack prior knowledge about their usability and associated issues [3, 39, 45].

RQ 3: How does the framework perform in real-world use cases?

The objective of this research question is to address the performance, customization, and demonstration of the framework in real-world experiments.

2 RELATED WORK

The utilization of smart wearable technologies and mobile health applications (mHealth apps) is prevalent and widespread in various domains such as fashion [41, 43, 66], gaming [17, 34, 46, 50, 71], industry, and safety [6, 10, 40, 49, 70]. However, the proliferated growth of using wearables and their associated mobile (health) apps within the healthcare industry has doubled in the last decade and the demand from the consumer sector continues to skyrocket [29]. These technologies are used for monitoring users' health and well-being. This means not just tracking physical activities like steps and calories burned but covering a wider range of uses like self-management of long-term conditions, preventive care, aftercare like diabetic monitoring [18, 64], rehabilitation [13, 20, 57], fall detection [8, 67], wound healing [26], and monitoring long-term illness symptoms [5, 22, 72]. Wearables have the potential to revolutionize healthcare and medical applications by improving access to patient data, remote patient monitoring, and health status tracking. By continuously collecting real-time health data, users can share it with healthcare providers and gain insights into health-related issues, such as patient behavior, disease onset, relapse indicators, symptom development, and progression over time [19, 48, 77]. These tools and technologies have transcended boundaries and are increasingly being adopted for clinical use, contributing to the improvement of patient assessment in terms of physical fitness and quality of life, as well as their application in preventive healthcare [23, 25, 31, 52].

In the field of wearable technologies and their associated mobile (health) applications there is a lack of well-defined attributes or methods that specifically cover the usability aspects such as comfort, durability, accessibility, safety, and wireless connectivity. Although established norms and regulations such as ISO 9241[30] and IEC 62366-1:2021-08 [11] define usability as how easy and functional a system is to use in a specific context as well as efficiency and satisfaction for specified users to achieve their intended goal. This assertion is further supported by previous research findings and empirical evidence in the field [56, 65]. These established standards and regulatory frameworks, recognized usability only as a result of use, emphasizing attributes like effectiveness, efficiency, and satisfaction. In addition to the outcome of use, the assessment of usability attributes beyond the immediate result also contributes to understanding the user's interaction with these technologies and their sustained adoption and acceptance. Regarding the initial stages of the dissertation, a scoping review was conducted to explore the literature relevant to various topics of interest. These topics include wearables, physiological data, mobile phones, mobile apps, telemedicine, usability, and user experience. The literature search aimed to gather information and insights on i) the types of wearables and/ or their combination with mHealth apps, ii) the frequency of use of wearables or together with their associated mobile application in a study, iii) medical use cases and medical data reported in studies and iv) attributes, methods and frameworks used to assess usability.

Usability assessment is critical to the success of wearable technology, as it increases productivity, reduces errors, training, and user support, and improves efficacy [27, 59]. However, the flexibility of wearable technology poses a challenge to assessing its usability in clinical settings, leading to a research gap. Additionally, the challenges related to usability, manageability, interoperability, availability, security, and privacy become increasingly pressing, highlighting the need for effective solutions and strategies [1, 42, 53, 58, 60]. The available literature on evaluation frameworks or guidelines for assessing the interaction between wearables and their associated apps is inadequate, as it lacks sufficient empirical data, essential usability attributes, methods, and standard evaluation guidelines.

In the realm of wearable technologies and mobile apps, design and development are crucial aspects. Over time, researchers have published various papers on different frameworks targeting specific purposes, such as design frameworks for wearables [2, 16, 21, 76], design frameworks for mHealth apps [24, 28, 32, 62], assessment of mHealth apps [7, 14, 15, 44], and even frameworks for the selection of wearables and mHealth apps for studies [36, 51, 61]. Moreover, the literature also reveals reviews on the usability evaluation of mHealth apps tailored to specific health conditions or patient populations, as demonstrated by previous studies [33, 55, 75]. While certain usability frameworks have been proposed, they primarily focus on apps alone and do not entirely fulfill the objective of evaluating the usability of the integration of wearables and their corresponding mobile apps. Furthermore, user-centered design approaches are commonly adopted for mHealth app development, while wearable design frameworks vary in focus, considering performance, social aspects, technology, and regulations. However, despite the abundance of design and development frameworks, interactive healthcare technologies still suffer from poor usability, hindering their widespread use. Additionally, there is a lack of comprehensive evaluation frameworks specifically designed for assessing the usability of wearable technologies and associated mobile apps.

Therefore, my objective is to bridge this research gap by proposing a novel framework designed to comprehensively evaluate the usability of wearable technologies and their associated mobile (health) apps. This framework aims to incorporate relevant usability attributes, as well as appropriate evaluation methods for effective assessment. Moreover, the framework seeks to consider additional factors that may impact usability measurements, such as acceptance and feasibility.

3 METHODOLOGY

In my dissertation, the objective is to investigate evaluation approaches and their shortcomings and propose a new evaluation framework. My research aims to address the gaps in current practices and to provide a comprehensive and effective methodology for assessing the usability of wearables and companion mobile (health) apps.

3.1 Analysis of current approach

In this study, I aim to investigate the predominant usability attributes, evaluation methods, and frameworks employed in the

assessment of wearables and mobile (health) apps. Through a comprehensive analysis of existing literature, I seek to identify the most frequently measured and evaluated factors in terms of usability. Additionally, examining current practices in research and industry will provide valuable insights into the key aspects that are considered when assessing the usability of wearables and mobile (health) apps.

a) Investigate the most commonly used and appropriate attributes and methods for evaluating wearables and companion apps.

i) Scope literature.

ii) Expert consultation and feedback.

b) Identify the usability attributes defined in the Medical Device Regulations (MDR) for evaluating usability and a thorough analysis of industry best practices.

c) Compare existing and proposed attributes to identify evaluation criteria differences and gaps.

3.2 Synthesis

After identifying potential improvements and advancements in usability evaluation methods from the literature, industry best practices, and expert feedback, these results inform the construction of a framework supporting the practical and effective evaluation of the usability of wearables and their companion apps. My goal is to create a framework that would be accessible and beneficial to both usability experts and non-experts. Therefore, the construction of the framework accessible and beneficial to both usability experts and non-experts will consider these factors:

a) Identify, select, and use the most appropriate usability attributes and evaluation techniques and tools.

i) Literature

ii) Expert feedback

b) Identify and analyze other factors that affect the measurement of usability and determine its necessity and purpose.

c) Introduce a graphical representation of the framework to represent familiarity, simplicity, and ease of interpretation.

3.3 Validation

For the validation of the usability evaluation framework for wearable technologies and companion mobile health apps, various established approaches can be utilized. These include comparative analysis, action learning, case studies, literature review, team collaboration, and expert feedback. By employing these techniques, the framework can be assessed and refined to ensure its effectiveness and relevance. Since there are no existing frameworks for evaluating the usability of these technologies, direct comparison with established frameworks like ISO 9241 is not feasible. Studies [35, 69] have shown the pros and cons of empirical testing and evidence-based practices including criteria for defining satisfactory evidence in evaluation frameworks. However, the use of various methods and guidelines leads to diverse outcomes based on the specific context of use and device interface. Such challenges can also be extended to the comparison of usability evaluation frameworks, as obtaining quantitative evidence of their effectiveness becomes intricate. Therefore, to validate and evaluate the proposed evaluation framework for the assessment of the usability of wearable

technologies and companion mobile health apps, we will follow a two-step process: i) expert feedback from industry and academia and ii) demonstrating the framework in real-world use cases in the form of case studies.

4 EVALUATION

4.1 Expert Feedback

For the validation and evaluation of the framework, experts specializing in human-computer interaction, medical informatics, and digital health, with a focus on usability and user experience, will be engaged. As suggested by Silva et al [68], expert review complements formative analysis before implementing it within a real-world use case with end-users. This lays the foundation for technology adaptation and acceptance by users and their context of use. A qualitative approach is being employed to validate the framework, using semi-structured interviews with experts. Additionally, along with the interview, the Repertory Grid Technique provides an effective approach for capturing subjective perceptions and interpretations related to a specific area of interest and minimizing the influence of interviewer bias [37]. Furthermore, these interviews seek to understand the necessary measures to assess the usability of wearables and their associated mobile (health) apps. Furthermore, I aim to clarify the rationale behind the inclusion and importance of individual usability attributes to ensure a comprehensive assessment. Based on expert feedback, the framework will be further refined before its utilization in real-world use cases to evaluate its usability.

4.2 Case Studies

Case studies serve as a means to provide explanations, descriptions, or explorations of everyday events by examining processes and practices in real-world settings. This includes interactions, their significance for a better understanding of the group under study, and any events or issues that occur in a real-life context [63]. To demonstrate the usability evaluation framework in a real-world setting, empirical evidence must be collected from real users through usability testing and user inquiries. In the scope of this research, the proposed framework will be applied in two distinct real-world use cases involving two clinics at the hospital (Neurology and Surgery). For example, they offer patients the opportunity to receive pre-/post-operative care, based on an assessment of their well-being. Such an initiative incorporates the use of wearable technologies and mobile apps, enabling teleconsultation and remote monitoring. By implementing such an approach, caregivers and physicians can closely monitor patients, and ensure their well-being. Consequently, such instances will serve as an opportunity to facilitate the comprehensive assessment of users' attitudes, acceptance, and experiences pertaining to the utilization of wearables and their companion mobile apps in everyday usage. We anticipate that the implementation of the framework across a broad spectrum of wearables and use cases will substantially enhance the validation of the framework. These case studies demonstrating the use of the framework with end-users will adapt the mixed method approach. Previous studies have demonstrated the advantages of implementing domain-specific frameworks in real-world scenarios to examine

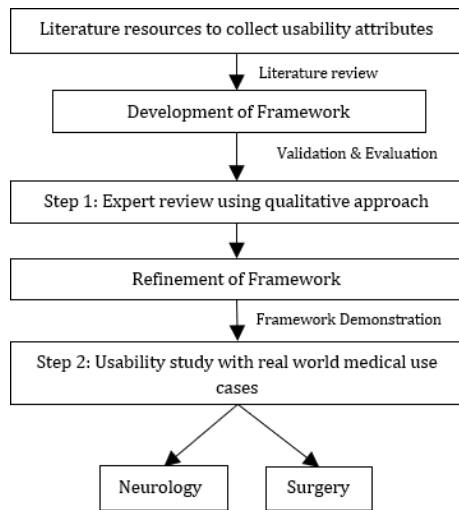


Figure 1: Two-step process of the validation and evaluation of the framework.

unique user behaviors and attitudes and their relationships to technology usage and the environment [9, 12, 73]. Figure 1 illustrates the research process.

4.2.1 Participants. The case studies will involve the recruitment of actual patients from the Neurology and Surgery clinics, at the University Medical Centre Mannheim. These clinics are actively engaged in clinical research projects that involve the utilization of various types of wearables such as smart rings and fitness trackers along with the associated mobile app, tailored specifically to their respective areas of specialization. Collaboration with the PIs from each clinic will facilitate the recruitment process. The usability study will take place in the Usability Lab of the Department of Biomedical Informatics, Centre for Preventive Medicine & Digital Health, Medical Faculty Mannheim of the University of Heidelberg following Ethical considerations adhering to the principles outlined in the Declaration of Helsinki [78]. Prior to the study, written consent will be obtained from all participants.

4.2.2 Procedure. Each use case will undergo a comprehensive usability testing session with respective patients from each clinic. The study will begin with an introductory session for participants, starting with the informed consent along with an overview of study objectives, an introduction to the wearable and mobile app, and explaining their usage. Participants will receive a wearable device such as a fitness/ activity tracker or a smart ring and a pre-installed mobile app. During the study, patients will be instructed to perform a series of specific tasks involving both the wearable device and its associated mobile application, allowing us to specifically assess their interaction with these technologies. Additionally, the usability study will enable us to methodically evaluate whether the designated methods effectively measure the intended usability attributes, consequently validating the suitability of both the usability attributes and the selected evaluation methods. Participants will have the option to withdraw at any stage of the study.

4.2.3 Data Collection. The usability study will employ a mixed-method approach to data collection, incorporating both quantitative and qualitative measures. These measures include task completion time, error rate, task success rate, user satisfaction, and semi-structured interviews. The selection of data collection methods however will depend on the specific usability attribute being measured depending on the type of wearable being used. Given the diverse range of usability attributes, careful selection of data collection methods is essential to ensure accuracy, precision, applicability, and relevance. The study results are aimed to assess the efficacy of the proposed usability attributes and methods.

4.2.4 Results / Outcomes. The case study results will serve as a process to further validate the framework's effectiveness in real-world use and assess the impact of usability attributes and methods on wearables and companion apps. Moreover, the assessment of the technologies' overall usability requires a thorough understanding of the target users, performance analysis of these technologies, evaluation of the technology's capabilities and limitations, and consideration of the environmental context of its usage. By achieving the objectives of this study, we can gain insights into user interactions with wearables and their companion mobile apps, thereby, enabling the development of a well-structured guideline, substantiated by empirical evidence, to validate the usability evaluation framework. Additionally, user feedback will inform usability, adoption, acceptance, and long-term usage. These findings will help refine the framework further and evaluate its overall performance.

5 (EXPECTED) CONTRIBUTION

In summary, this PhD research is expected to make the following contributions. The proposed framework is designed to

- Address the lack of appropriate usability attributes and guidelines for the assessment of wearables and mobile applications.

- Provide a foundation for evaluating usability using suitable attributes and methods for such a combination of devices/ technologies.

- Visual representation of the framework to provide accessibility and approachability for a wide range of non-expert users.

- Accessible and adaptable for usability experts and non-experts in real-world use cases.

6 STAGES OF RESEARCH

This research is still in its early stages. The ongoing process involves a comprehensive literature review to identify existing usability attributes, methods, and frameworks, as well as relevant case studies and the prevalence of wearables in studies. Currently, the completion of the scoping review paper is underway. In addition, preparations for the case studies have been made in collaboration with the PIs. Moreover, approval from the ethics committee has been obtained to conduct the study with real patients.

The main challenge is identifying the appropriate usability attributes and selecting the most suitable methods for evaluating wearables and their companion apps, considering the multitude of approaches available for measuring each attribute. Furthermore, the objective of this research study is to also align the evaluation standards with the usability attributes specified in the Medical Device

Regulation (MDR). However, the process of identifying, investigating, and comparing the proposed attributes with the usability attributes listed in the MDR may present certain challenges and limitations during the validation of the framework. Additionally, the adoption and acceptance of these technologies by clinicians and patients in clinical settings also present significant barriers. Further challenges to enhance the accessibility and approachability of the framework for a diverse set of non-expert users require careful consideration and resolution. These challenges are outlined to serve as open topics for discussion.

7 CONCLUSIONS

The objective of my doctoral research is to develop a practical and applicable usability framework for evaluating the usability of wearable technologies and their companion mobile applications in the domain of healthcare. This framework aims to be accessible not only to usability experts but also to non-experts, including clinicians and researchers. Hence, the interaction with experts in Human-Computer Interaction and wearable technology at the UbiComp/ISWC conference will enrich and advance my research.

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Preetha Moorthy has been a PhD student at the Medical Faculty Mannheim of Heidelberg University in Germany since October 2022. She is part of the research group Centre for Innovative After-care (Zentrum für innovative Versorgung- ZIV). She is particularly interested in research on the adaptation and acceptance of wearable technologies and mobile (health) applications within the healthcare sector especially in clinical settings. Her advisor is Prof. Dr. Till Nagel, Professor of Visual Analytics at the Mannheim University of Applied Sciences, and an associated member of the Medical Faculty Mannheim of Heidelberg University. She plans to finalize her dissertation around 2025.

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